

Cybersecurity Palo Alto Basic URL Filtering Set-up



By Rui Yang
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Purpose:

The Purpose of this lab is to set up URL Filtering for an elementary school in the US. The goal is to restrict the students from accessing non-school related content on school computers and in school in general. This allows students to focus on school and learning instead of becoming distracted from online distractions. It also ensures the safety of private information such as account details.

Background Information:

Palo Alto Networks:

Palo Alto Networks is a global leader in cybersecurity, founded in 2005 by Nir Zuk. Based in Santa Clara, California, the company initially disrupted the market with its Next Generation Firewalls (NGFWs) and has since expanded into a comprehensive, multinational platform for network security, cloud security (Prisma Cloud), and security operations (Cortex). It is focused on becoming a central platform for security by consolidating data and analytics to protect against digital threats. The company offers zero trust network security, cloud security, security analytics and automation, threat intelligence, security consulting, and secure access service edge.

Palo Alto PA-410:

It is part of the Palo Alto Networks PA-400 Series Next-Generation Firewalls. It brings ML-Powered NGFW capabilities to distributed enterprise branch offices, retail locations, and midsize businesses. It's the world's first ML-Powered Next-Generation Firewall, which enables you to prevent unknown threats, see and secure everything including the Internet of Things (IoT) and reduce errors with automatic policy recommendations. It is controlled by the same Pan-OS that controls and runs all Palo Alto NGFWs. PAN-OS natively classifies all traffic, inclusive of applications, threats, and content, and then ties that traffic to the user regardless of location or device type. Some of its functions include but not limited to: Enables SD-WAN Functionality, Detects and Prevents Advanced Threats with Cloud-Delivered Security Services, Offers AI-Powered Unified Management and Operations with Strata Cloud Manager, and Identifies and Categorizes All Applications, on All Ports, All the Time, with Full Layer 7 Inspection.

URL Filtering:

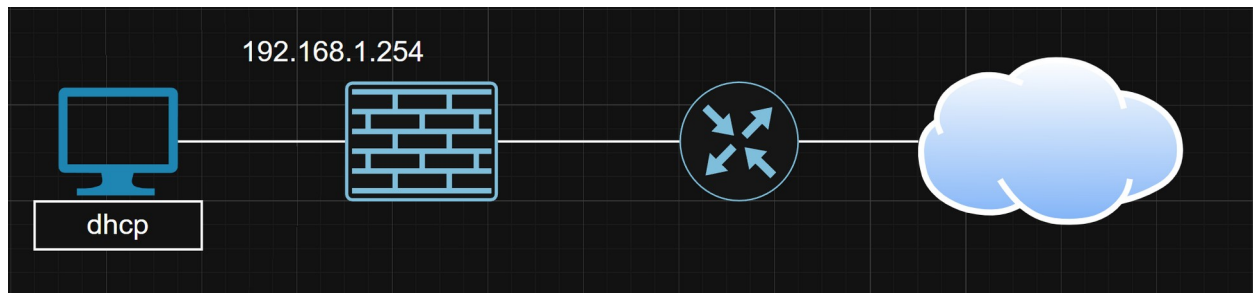
URL Filtering is a security feature on Palo Alto Networks firewalls that

controls web traffic based on website categories. Its main purposes are to protect users from threats (malware sites, phishing, command-and-control servers), enforce corporate policies (e.g., block adult content, gambling, or streaming during work hours), manage productivity (restrict social media, job search, shopping), and prevent credential theft (block users from submitting corporate logins to risky or untrusted sites).

Lab Summary:

Successfully created the outbound rules which restricted most of the non-school related content such as adult content, gambling, weaponry, and AI assisted writing tools, as well as parked and high-risk websites that could steal the school information.

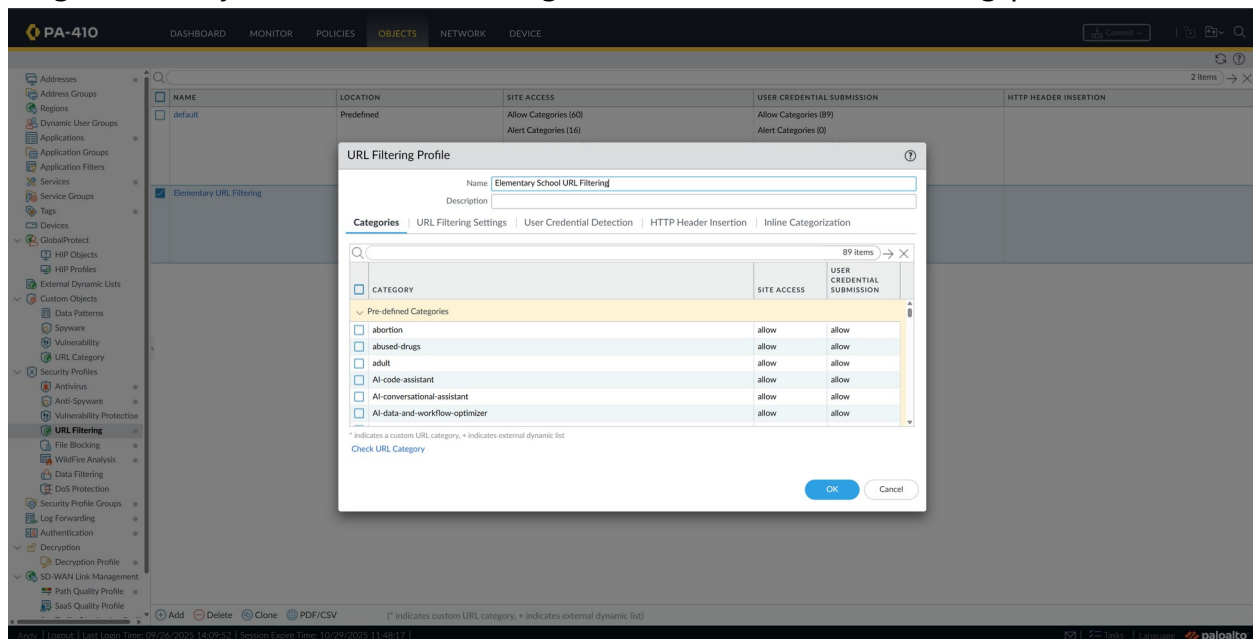
Network Diagram:



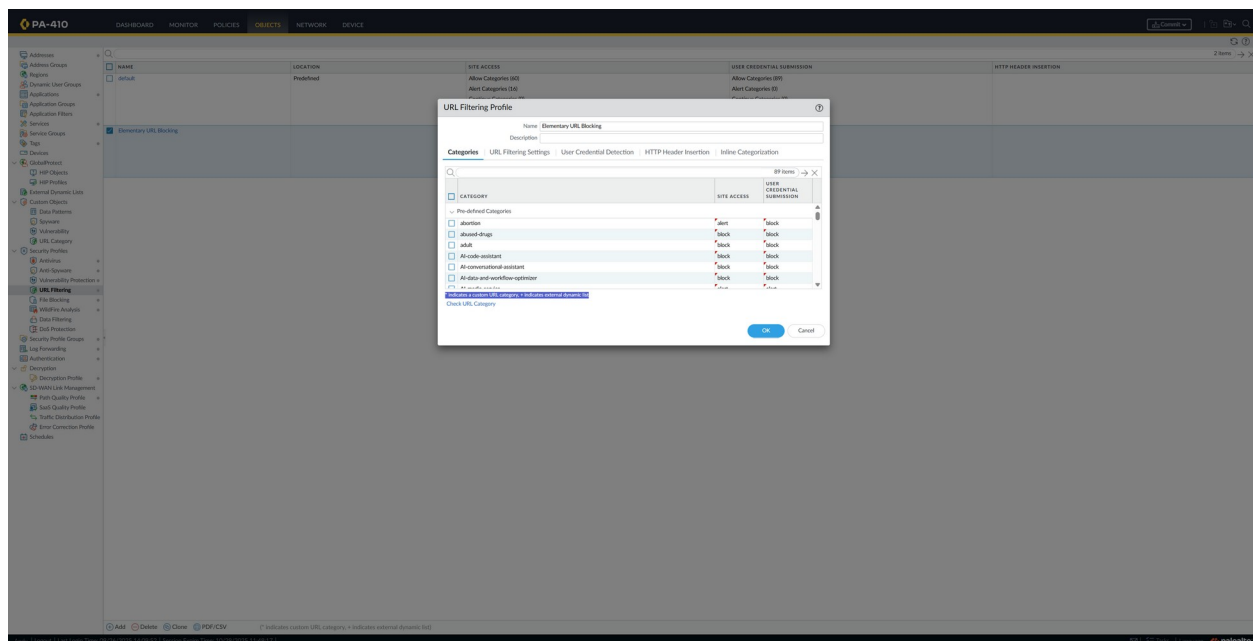
Steps and Configuration:

For this lab, it requires a basic SoHo Configuration.

navigate to Objects -> URL Filtering -> add a new URL Filtering profile.



Manually set the site access and user credential submissions to desired settings - This lab was made for an elementary school; The restrictions are harsher.



URL Filtering Profile

Name

Elementary URL Blocking

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☒	AI-media-service	alert	alert
☐	AI-meeting-assistant	block	block
☐	AI-platform-service	block	block
☐	AI-website-generator	block	block
☐	AI-writing-assistant	block	block
☐	alcohol-and-tobacco	alert	block
☐	artificial-intelligence	block	block

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

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HTTP Header Insertion

Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☒	auctions	alert	block
☐	business-and-economy	alert	alert
☐	command-and-control	block	block
☐	compromised-website	block	block
☐	computer-and-internet-info	alert	alert
☐	content-delivery-networks	alert	alert
☐	copyright-infringement	block	block

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

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Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☒	high-risk	block	block
☐	home-and-garden	allow	allow
☐	hunting-and-fishing	allow	allow
☐	insufficient-content	block	block
☐	internet-communications-and-telephony	alert	alert
☐	internet-portals	alert	alert
☐	job-search	block	block
☐	legal	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

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URL Filtering Settings

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HTTP Header Insertion

Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☒	low-risk	alert	alert
☐	malware	block	block
☐	marijuana	block	block
☐	medium-risk	block	block
☐	military	alert	alert
☐	motor-vehicles	alert	alert
☐	music	alert	alert
☐	newly-registered-domain	block	block

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

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?

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Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☒	news	alert	alert
☐	not-resolved	block	block
☐	nudity	block	block
☐	online-storage-and-backup	alert	alert
☐	parked	block	block
☐	peer-to-peer	block	block
☐	personal-sites-and-blogs	alert	allow
☐	philosophy-and-political-advocacy	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

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Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☒	phishing	block	block
☐	private-ip-addresses	block	block
☐	proxy-avoidance-and-anonymizers	block	block
☐	questionable	block	block
☐	ransomware	block	block
☐	real-estate	alert	block
☐	real-time-detection	alert	alert
☐	recreation-and-hobbies	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

URL Filtering Profile?

Name

Elementary URL Filtering

Description

- Categories
- URL Filtering Settings
- User Credential Detection
- HTTP Header Insertion
- Inline Categorization

89 items

→

×

☐ CATEGORY ^	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐ reference-and-research	alert	alert
☐ religion	alert	alert
☒ remote-access	alert	alert
☐ scanning-activity	block	block
☐ search-engines	alert	alert
☐ sex-education	block	block
☐ shareware-and-freeware	alert	allow
☐ shopping	alert	alert

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[Check URL Category](#)

OK

Cancel

URL Filtering Profile?

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Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☒	social-networking	alert	alert
☐	society	alert	alert
☐	sports	alert	alert
☐	stock-advice-and-tools	alert	alert
☐	streaming-media	alert	alert
☐	swimsuits-and-intimate-apparel	alert	alert
☐	training-and-tools	alert	alert
☐	translation	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

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User Credential Detection

HTTP Header Insertion

Inline Categorization

Q

89 items

→

×

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
☐	training-and-tools	alert	alert
☒	translation	alert	alert
☐	travel	alert	alert
☐	unknown	block	block
☐	weapons	block	block
☐	web-advertisements	alert	alert
☐	web-based-email	alert	alert
☐	web-hosting	alert	alert

* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

Cancel

Enable local inline categorization:

URL Filtering Profile

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Elementary URL Filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

☒ Enable local inline categorization

☐ Enable cloud inline categorization

Exceptions

☐ CUSTOM URL CATEGORY/EDL

+ Add

- Delete

OK

Cancel

Enable Safe Search Enforcement:

URL Filtering Profile

Name

Elementary URL Filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

☒ Log container page only

☒ Safe Search Enforcement

HTTP Header Logging

☒ User-Agent

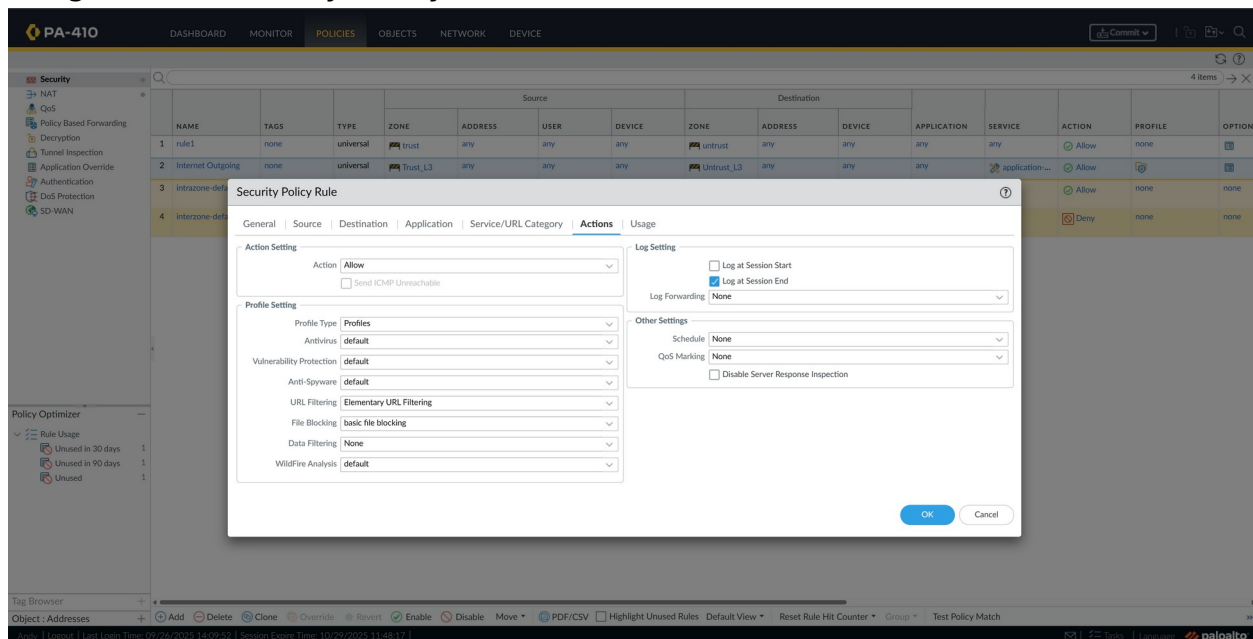
☐ Referer

☒ X-Forwarded-For

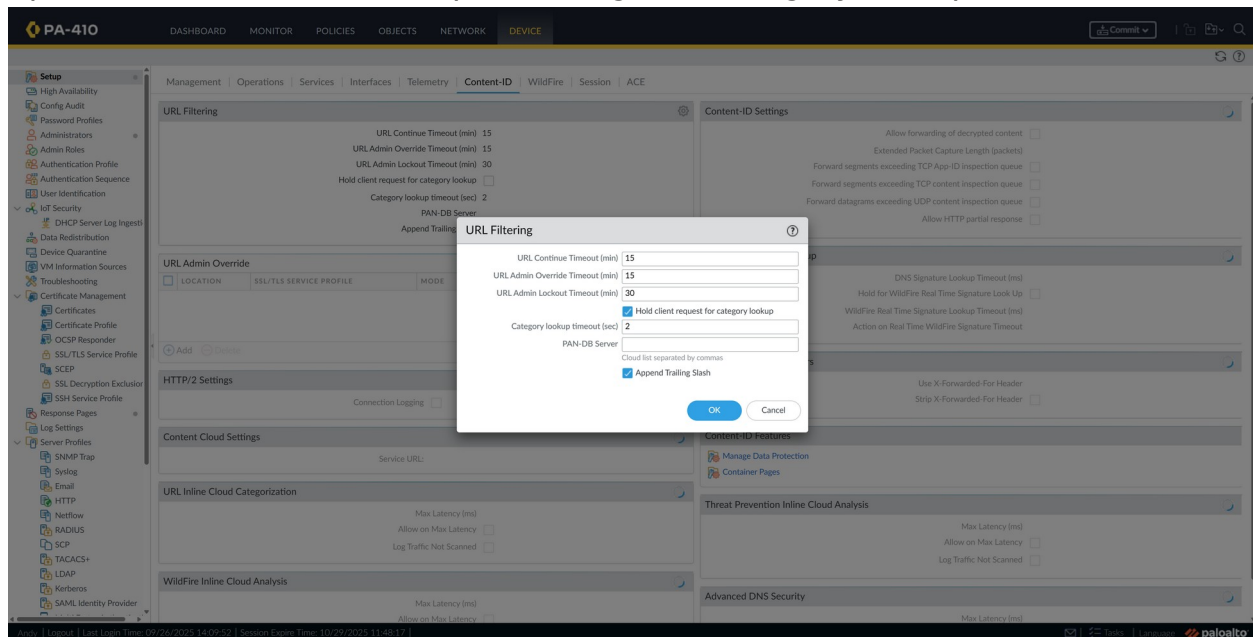
OK

Cancel

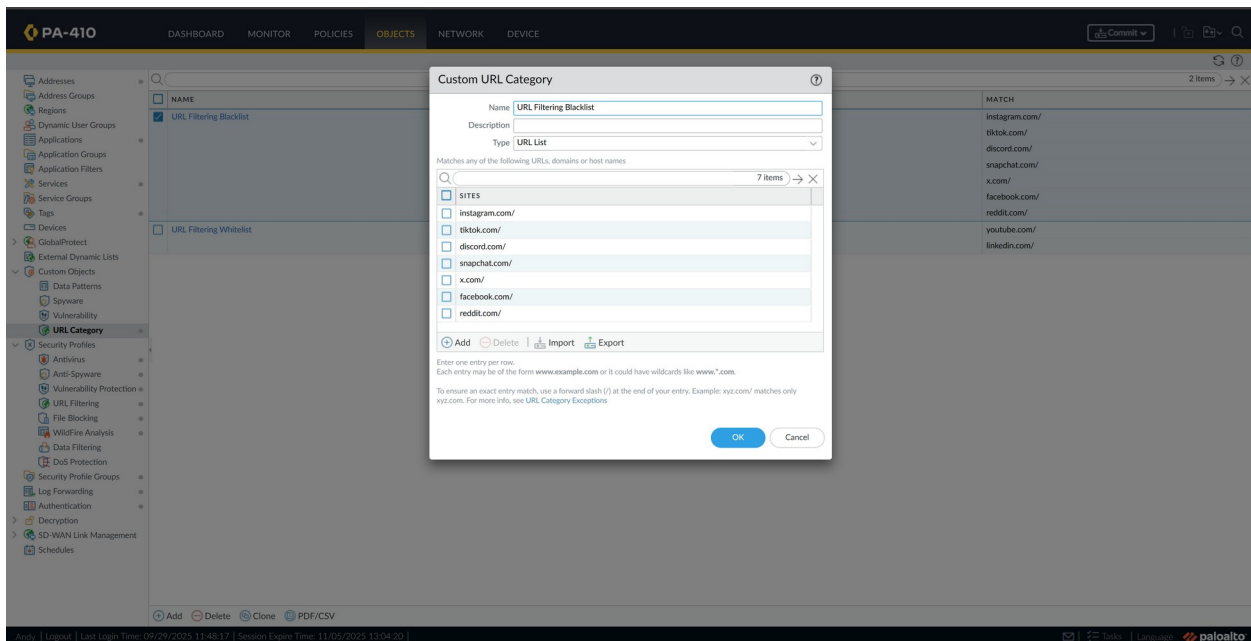
Configure the Security Policy Rules:



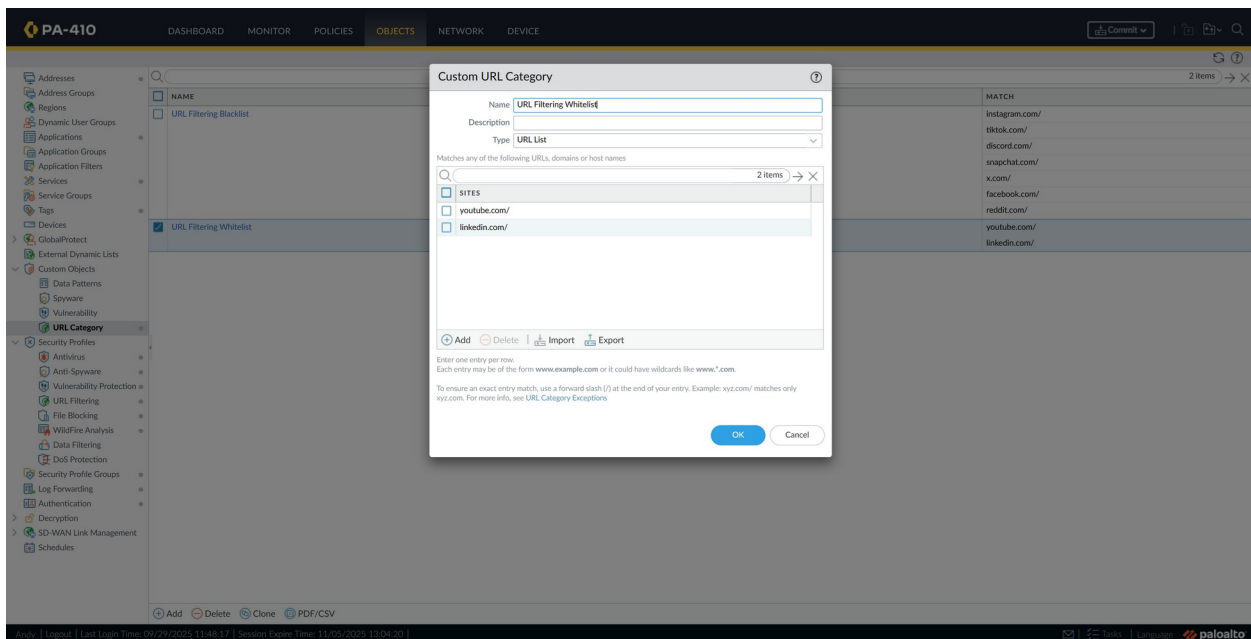
Enable “Hold client request for category lookup” to block clients from requests while the firewall is performing URL category lookups.



Create a custom URL blacklist to add all blocked sites to this blacklist:



Now also make one for whitelisting blocked sites that are falsely categorized or ones administrators wish to unblock:



Add rules for these custom categories inside the URL Filtering Profile:

URL Filtering Profile

Name

Elementary URL Filtering

Description

Categories

URL Filtering Settings

User Credential Detection

HTTP Header Insertion

Inline Categorization

91 items

☐	CATEGORY	SITE ACCESS	USER CREDENTIAL SUBMISSION
Custom URL Categories			
☐	URL Filtering Blacklist *	block	block
☒	URL Filtering Whitelist *	alert	alert
Pre-defined Categories			
☐	abortion	alert	block
☐	abused-drugs	block	block
☐	adult	block	block

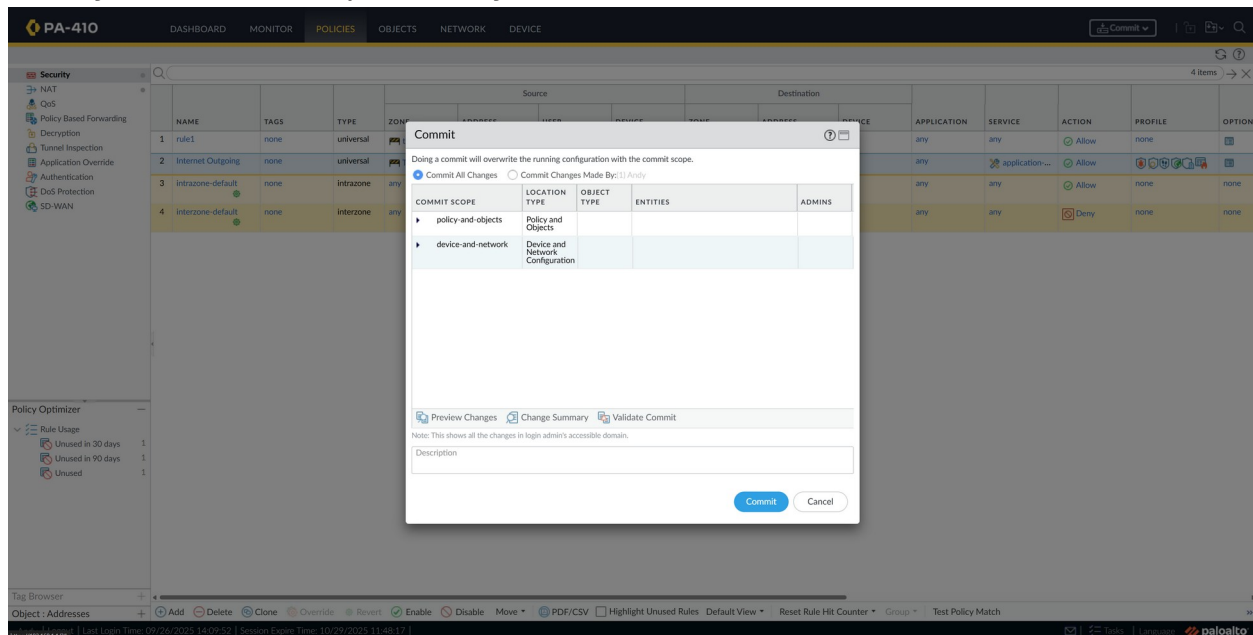
* indicates a custom URL category, + indicates external dynamic list

[Check URL Category](#)

OK

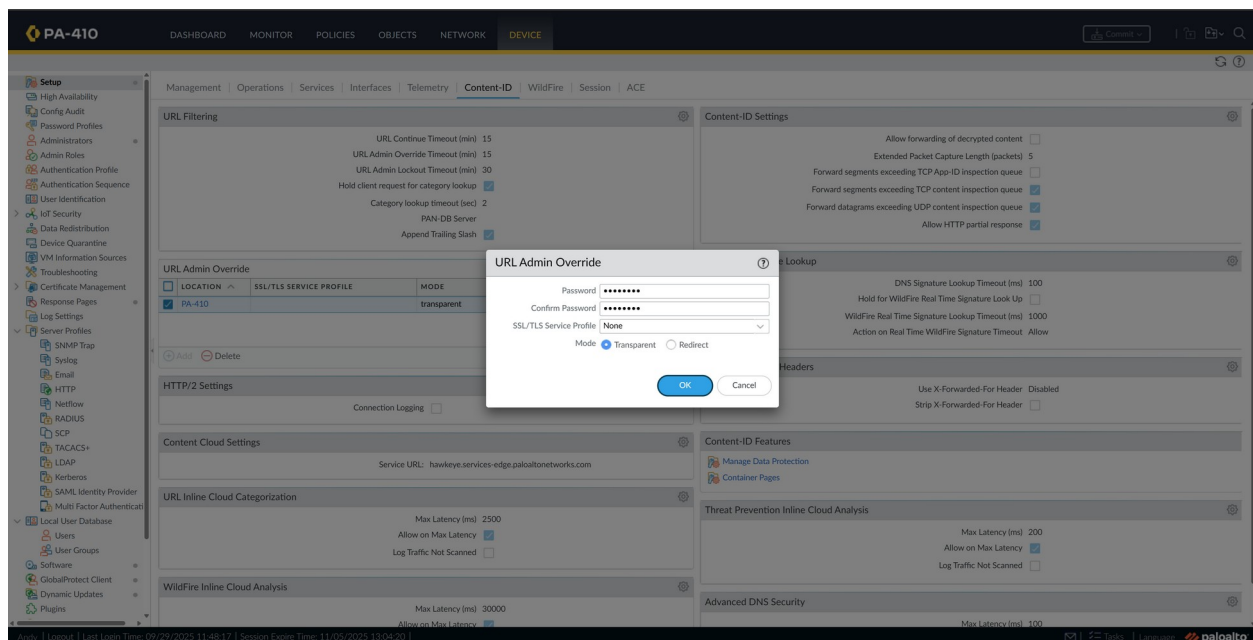
Cancel

Commit All Changes for it apply (the commit screens will vary depends on where you committed previously)



Optional: Configure and enable URL Admin Override:

Configure a custom URL Admin Override Password in Setup -> URL Admin Override ->Add:



And then go back to the URL Filtering Profile and change all the sites that are being blocked to override, which then you can enter an admin password that was just set to access the restricted websites.

91 items → ×			
☐ CATEGORY ^	SITE ACCESS	USER CREDENTIAL SUBMISSION	
Custom URL Categories			
☐ URL Filtering Blacklist *	block	block	
☐ URL Filtering Whitelist *	alert	alert	
Pre-defined Categories			
☐ abortion	alert	block	
☐ abused-drugs	override	block	
☒ adult	block	block	
☐ AI-code-assistant	override	block	
☐ AI-conversational-assistant	override	block	
☐ AI-data-and-workflow-optimizer	override	block	
☐ AI-media-service	alert	alert	
☐ AI-meeting-assistant	override	block	
☐ AI-platform-service	override	block	
☐ AI-website-generator	override	block	
☐ AI-writing-assistant	override	block	
☐ alcohol-and-tobacco	alert	block	
☐ artificial-intelligence	override	block	
☐ auctions	alert	block	
☐ business-and-economy	alert	alert	
☐ command-and-control	override	block	
☐ compromised-website	override	block	
☐ computer-and-internet-info	alert	alert	
☐ content-delivery-networks	alert	alert	
☐ copyright-infringement	override	block	
☐ cryptocurrency	override	block	
☐ dating	override	block	
☐ dynamic-dns	alert	alert	
☐ educational-institutions	alert	alert	
☐ encrypted-dns	override	block	
☐ entertainment-and-arts	alert	alert	
☐ extremism	override	block	
☐ file-converter	alert	allow	
☐ financial-services	override	block	
☐ gambling	override	block	
☐ games	alert	alert	
☐ government	alert	alert	
☐ grayware	override	block	

☐	hacking	override	block
☐	health-and-medicine	alert	alert
☐	high-risk	override	block
☐	home-and-garden	alert	alert
☐	hunting-and-fishing	alert	alert
☐	insufficient-content	override	block
☐	internet-communications-and-telephony	alert	alert
☐	internet-portals	alert	alert
☐	job-search	override	block
☐	legal	alert	alert
☐	low-risk	alert	alert
☐	malware	override	block
☐	marijuana	override	block
☐	medium-risk	override	block
☐	military	alert	alert
☐	motor-vehicles	alert	alert
☐	music	alert	alert
☐	newly-registered-domain	override	block
☐	news	alert	alert
☐	not-resolved	override	block
☐	nudity	block	block
☐	online-storage-and-backup	alert	alert
☐	parked	override	block
☐	peer-to-peer	override	block
☐	personal-sites-and-blogs	alert	allow
☐	philosophy-and-political-advocacy	alert	alert
☐	phishing	block	block
☐	private-ip-addresses	override	block
☐	proxy-avoidance-and-anonymizers	block	block
☐	questionable	override	block
☐	ransomware	block	block
☐	real-estate	alert	block
☐	real-time-detection	alert	alert
☐	recreation-and-hobbies	alert	alert
☐	reference-and-research	alert	alert
☐	religion	alert	alert
☐	remote-access	alert	alert
☐	scanning-activity	override	block
☐	search-engines	alert	alert
☐	sex-education	block	block
☐	shareware-and-freeware	alert	allow
☐	shopping	alert	alert

☐	social-networking	alert	alert
☐	society	alert	alert
☐	sports	alert	alert
☐	stock-advice-and-tools	alert	alert
☐	streaming-media	alert	alert
☐	swimsuits-and-intimate-apparel	alert	alert
☐	training-and-tools	alert	alert
☐	translation	alert	alert
☐	travel	alert	alert
☐	unknown	override	block
☐	weapons	block	block
☐	web-advertisements	alert	alert
☐	web-based-email	alert	alert
☐	web-hosting	alert	alert

Then commit changes:

Commit

?

Doing a commit will overwrite the running configuration with the commit scope.

☒ Commit All Changes

☐ Commit Changes Made By:(1) Andy

COMMIT SCOPE	LOCATION TYPE	OBJECT TYPE	ENTITIES	ADMINS
▶ policy-and-objects	Policy and Objects			

Preview Changes

Change Summary

Validate Commit

Note: This shows all the changes in login admin's accessible domain.

Description

Commit

Cancel

Issues:

The different categories within the URL blocking built into Palo Alto devices were relatively vague based off the name, so I had to reference the official Palo Alto documentation and find out information about what each section meant, as well the URL filtering Settings and HTTP Header logging settings.

Conclusion:

This lab created a basic rough setup of a filter for an elementary school. Other restricted sites can be added to the rules as well if necessary. It required some research through the Palo Alto Official Documentation, but it was relatively simple to set up. This allowed me to get more used to the Palo Alto PA-410 Web GUI as well as learned how to set up URL filtering on a PA-410. [Palo Alto URL Filtering Guide.](#)